

Problem 1b: The Expenditure Function (Starting from Scratch)

ECO 310 - For People Who Skipped Lecture

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1 What Are We Doing Again?

Part (a) asked: "What should you buy to reach utility u^* cheaply?"

Part (b) asks: "Okay, now how much does that cost?"

That's it! The expenditure function is just the cost of what you found in part (a).

2 Super Basic Definition

Expenditure Function

Expenditure function = How much money you need to spend to reach utility level u^*
It's literally: (price of good 1) $\times$ (amount of good 1) + (price of good 2) $\times$ (amount of good 2)

Formula:

$$e(p_1, p_2, u^*) = p_1 \cdot x_1^c + p_2 \cdot x_2^c$$

where x_1^c and x_2^c are from your answer to part (a).

3 Starting with Examples (Coffee and Tea)

Let's use actual numbers first, then write the general formula.

3.1 Example 1: Coffee is Cheaper

Given:

- Coffee costs \$3 per cup ($p_1 = 3$)
- Tea costs \$5 per cup ($p_2 = 5$)
- You want utility 10 ($u^* = 10$)
- Utility function: $u = \text{coffee} + \text{tea}$

From Part (a): Since coffee is cheaper ($3 < 5$), you buy:

$$(x_1^c, x_2^c) = (10, 0)$$

That means: 10 coffees, 0 teas.

Part (b) - Calculate expenditure:

$$\begin{aligned} e(3, 5, 10) &= (\text{price of coffee}) \times (\text{coffees bought}) + (\text{price of tea}) \times (\text{teas bought}) \\ &= 3 \times 10 + 5 \times 0 \\ &= 30 + 0 \\ &= \$30 \end{aligned}$$

Answer: It costs \$30 to reach utility level 10.

3.2 Example 2: Tea is Cheaper

Given:

- Coffee costs \$5 per cup ($p_1 = 5$)
- Tea costs \$2 per cup ($p_2 = 2$)
- You want utility 10 ($u^* = 10$)

From Part (a): Since tea is cheaper ($2 < 5$), you buy:

$$(x_1^c, x_2^c) = (0, 10)$$

That means: 0 coffees, 10 teas.

Part (b) - Calculate expenditure:

$$\begin{aligned} e(5, 2, 10) &= 5 \times 0 + 2 \times 10 \\ &= 0 + 20 \\ &= \$20 \end{aligned}$$

Answer: It costs \$20 to reach utility level 10.

3.3 Example 3: Different Utility Target

Given:

- Coffee costs \$3 per cup ($p_1 = 3$)
- Tea costs \$5 per cup ($p_2 = 5$)
- You want utility 25 ($u^* = 25$)

From Part (a): Since coffee is cheaper, you buy:

$$(x_1^c, x_2^c) = (25, 0)$$

Part (b) - Calculate expenditure:

$$\begin{aligned} e(3, 5, 25) &= 3 \times 25 + 5 \times 0 \\ &= 75 + 0 \\ &= \$75 \end{aligned}$$

Answer: It costs \$75 to reach utility level 25.

4 See the Pattern?

Look at the examples:

- When coffee is cheaper (\$3 vs \$5): $\text{cost} = (\$3) \times (\text{utility target})$
- When tea is cheaper (\$2 vs \$5): $\text{cost} = (\$2) \times (\text{utility target})$

Pattern: $\text{cost} = (\text{cheaper price}) \times (\text{utility target})$

5 Writing the General Formula

Now let's do this without specific numbers. Use p_1 , p_2 , and u^* as variables.

5.1 Recall Part (a) Answer

From part (a), compensated demand is:

$$(x_1^c, x_2^c) = \begin{cases} (u^*, 0) & \text{if } p_1 < p_2 \\ (0, u^*) & \text{if } p_1 > p_2 \end{cases}$$

5.2 Calculate Expenditure for Each Case

The expenditure function is: $e(p_1, p_2, u^*) = p_1 \cdot x_1^c + p_2 \cdot x_2^c$

Case 1: If $p_1 < p_2$ (good 1 is cheaper)

From part (a): $(x_1^c, x_2^c) = (u^*, 0)$

Plug into expenditure formula:

$$\begin{aligned} e(p_1, p_2, u^*) &= p_1 \cdot u^* + p_2 \cdot 0 \\ &= p_1 u^* \end{aligned}$$

Case 2: If $p_1 > p_2$ (good 2 is cheaper)

From part (a): $(x_1^c, x_2^c) = (0, u^*)$

Plug into expenditure formula:

$$\begin{aligned} e(p_1, p_2, u^*) &= p_1 \cdot 0 + p_2 \cdot u^* \\ &= p_2 u^* \end{aligned}$$

Case 3: If $p_1 = p_2$ (same price)

From part (a): any combination works, like $(u^*, 0)$ or $(0, u^*)$ or $(u^*/2, u^*/2)$

But they all cost the same:

$$e(p_1, p_2, u^*) = p_1 u^* = p_2 u^* \quad (\text{since } p_1 = p_2)$$

6 Final Answer to Part (b)

Answer: Expenditure Function (Piecewise)

$$e(p_1, p_2, u^*) = \begin{cases} p_1 u^* & \text{if } p_1 < p_2 \\ p_2 u^* & \text{if } p_1 > p_2 \\ p_1 u^* = p_2 u^* & \text{if } p_1 = p_2 \end{cases} \quad (1)$$

6.1 Even Simpler Form

You can write this more compactly as:

Answer: Expenditure Function (Compact)

$$e(p_1, p_2, u^*) = \min\{p_1, p_2\} \cdot u^* \quad (2)$$

Translation: Minimum cost = (whichever price is lower) $\times$ (target utility)
Both forms are correct! The compact form is cleaner.

7 Verify with Our Examples

Let's check that our formula works:

7.1 Example 1 Check

Coffee \$3, tea \$5, $u^* = 10$

Using formula: $e(3, 5, 10) = \min\{3, 5\} \times 10 = 3 \times 10 = 30 \checkmark$

7.2 Example 2 Check

Coffee \$5, tea \$2, $u^* = 10$

Using formula: $e(5, 2, 10) = \min\{5, 2\} \times 10 = 2 \times 10 = 20 \checkmark$

7.3 Example 3 Check

Coffee \$3, tea \$5, $u^* = 25$

Using formula: $e(3, 5, 25) = \min\{3, 5\} \times 25 = 3 \times 25 = 75 \checkmark$

Perfect! Our formula works.

8 Why Does This Make Sense?

Intuition

Question: What's the cheapest way to get u^* units of utility?

Answer: With perfect substitutes, buy only the cheaper good.

Cost: (cheaper price) $\times$ (how many units you need) = (cheaper price) $\times u^*$

This is exactly what $e(p_1, p_2, u^*) = \min\{p_1, p_2\} \cdot u^*$ says!

9 Step-by-Step Process for Part (b)

1. Look at your answer from part (a) - what bundle do you buy in each case?
2. For each case, calculate total cost: $p_1x_1 + p_2x_2$
3. Simplify the expressions

4. Write as piecewise function OR use $\min\{p_1, p_2\}$ notation
5. Done!

10 Common Mistakes to Avoid

- **Don't** forget to multiply by u^* - the cost depends on how much utility you want!
- **Don't** average the prices - you buy only the cheaper good, not a mix
- **Don't** use m (income) - expenditure function uses u^* (target utility), not budget

11 Summary

TL;DR

Part (a): Find what to buy $\rightarrow (u^*, 0)$ if good 1 cheaper, $(0, u^*)$ if good 2 cheaper

Part (b): Calculate the cost $\rightarrow e(p_1, p_2, u^*) = \min\{p_1, p_2\} \cdot u^*$

Intuition: Buy the cheaper good, so cost = (cheaper price) $\times$ (target utility)